Diabetes and the Mediterranean diet: a beneficial effect of oleic acid on insulin sensitivity, adipocyte glucose transport and endothelium-dependen...

Abnormalities in endothelial function may be associated with increased cardiovascular risk in diabetic patients. We examined the effect of an oleic-acid-rich diet on insulin resistance and endothelium-dependent vasoreactivity in type 2 diabetes. Eleven type 2 diabetic patients were changed from their usual linoleic-acid-rich diet and treated for 2 months with an oleic-acid-rich diet. Insulin-mediated glucose transport was measured in isolated adipocytes. Fatty acid composition of the adipocyte membranes was determined by gas-liquid chromatography and flow-mediated endothelium-dependent and -independent vasodilatation were measured in the superficial femoral artery at the end of each dietary period. There was a significant increase in oleic acid and a decrease in linoleic acid on the oleic-acid-rich diet ($p<0.0001$). Diabetic control was not different between the diets, but there was a small but significant decrease in fasting glucose/insulin on the oleic-acid-rich diet. Insulin-stimulated (1 ng/ml) glucose transport was significantly greater on the oleic- acid-rich diet (0.56+/-0.17 vs. 0.29+/-0.14 nmol/10(5) cells/3 min, $p<0.0001$). Endothelium-dependent flow-mediated vasodilatation (FMD) was significantly greater on the oleic-acid-rich diet (3.90+/-0.97% vs. 6.12+/-1.36% $p<0.0001$). There was a significant correlation between adipocyte membrane oleic/linoleic acid and insulin-mediated glucose transport ($p<0.001$) but no relationship between insulin-stimulated glucose transport and change in endothelium-dependent FMD. There was a significant positive correlation between adipocyte membrane oleic/linoleic acid and endothelium-dependent FMD ($r=0.61$, $p<0.001$). Change from polyunsaturated to monounsaturated diet in type 2 diabetes reduced insulin resistance and restored endothelium-dependent vasodilatation, suggesting an explanation for the anti-atherogenic benefits of a Mediterranean-type diet.